



Term Evaluation Theory (Even) Semester Regular Examination May 2026

Roll no. 2294038

Name of the Course: B.Tech C.S.E

Semester: VIII

Name of the Paper: Disaster Management

Paper Code: TDM 881

Time: 1.5 hour

Maximum Marks: 50

Note:

- (i) Answer all the questions by choosing any one of the sub-questions
- (ii) Each question carries 10 marks.

Q1. (10 Marks)
a. Describe the causes, distribution, and hazardous effects of earthquakes. Discuss their environmental, social, and economic impacts; and explain the mitigation and preparedness measures adopted in India to reduce earthquake risk. [CO1]

OR

b. Discuss the phenomenon of cloud bursts and glacial hazards with reference to their occurrence in Uttarakhand. What disaster preparedness measures are needed? [CO1]

Q2. (10 Marks)
a. Explain the causes and characteristics of man-induced disasters such as water logging, subsidence, and ground water depletion. Discuss their social, economic, and environmental consequences. [CO1]

OR

b. Compare and contrast natural disasters and man-induced disasters with suitable examples. Discuss their causes, impacts, and characteristics, and explain how mitigation and management strategies differ for both types of disasters [CO1]

Q3. (10 Marks)
a. Discuss the hazards associated with volcanic activity and mud volcanoes. Identify the major volcanic hazard zones in India and explain their environmental and socio-economic impacts. [CO1]

OR

b. Identify and evaluate the major natural disaster-prone areas in India with special reference to Uttarakhand. Suggest region-specific mitigation measures. [CO2]

Q4. (10 Marks)
a. Explain sustainable development and its role in disaster mitigation. How does unsustainable development increase disaster risk? [CO2]

OR

b. Discuss the differential impacts of disasters on vulnerable groups such as women, children, and economically weaker sections. [CO2]

Q5. (10 Marks)
a. Critically analyze the role of community participation in disaster mitigation and preparedness. Discuss the initiatives and practices that have been effective in Indian villages for reducing disaster risks and improving resilience. [CO2]

OR

b. Examine the roles and responsibilities of State and Central Governments in disaster management. Discuss how coordination among government agencies, local authorities, NGOs, and communities strengthens disaster risk reduction and response mechanisms. [CO2]